NATPOWER DIGITAL × CMA CGM · ALBERT CANAL

# Trip report · Antwerp to Meerhout and back

Scenario 2 · Tender at the berth · Dry summer · CMA CGM e-barge, 135 m, two 2.9 MWh ZESpacks

**138 km**
route length, 4 locks

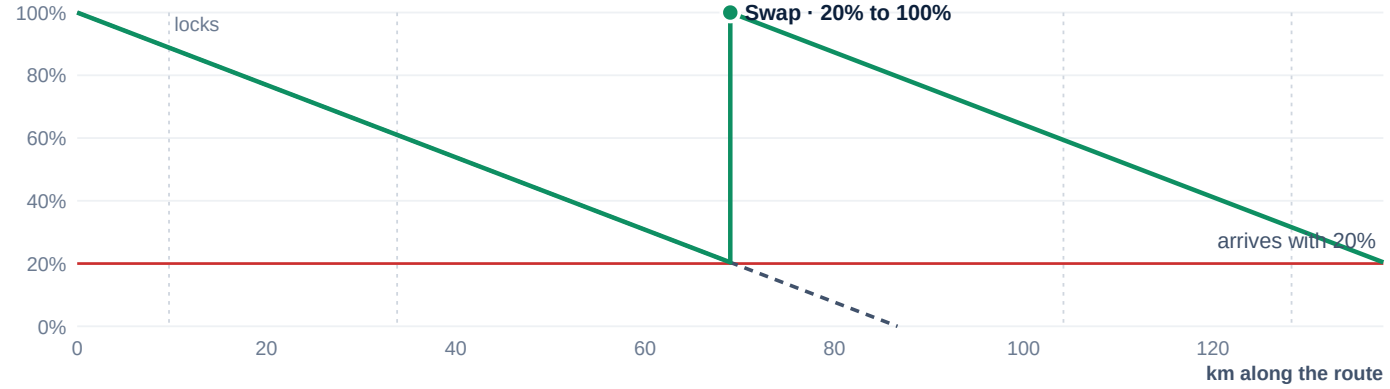
**24 h 50**
total time on the water, with group lockage

**8.3 MWh**
energy at the propeller

**4.1 t**
CO2 saved against a diesel barge

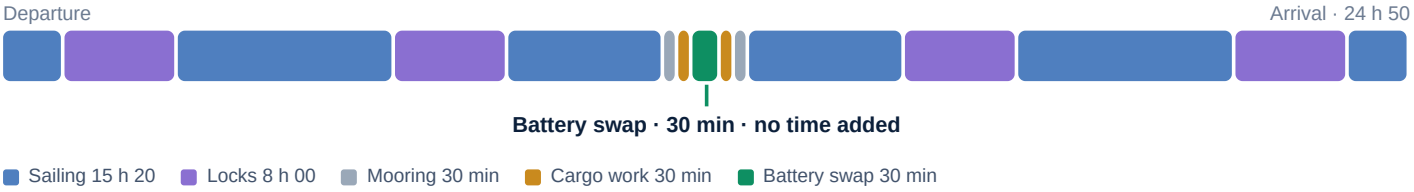
## Battery charge along the route

Shown as the share of usable energy in the two packs. Without a swap the packs reach the 20% reserve at km 69 and run flat before the end of the trip. The swap at the BCTN Meerhout berth, during cargo work puts them back to 100%.




## The day on one line

The battery tender comes alongside while the cranes work the boxes; the 30 minute swap fits inside the call. In a dry summer the Meuse runs low and ships are locked through in groups, so each lock can take two to four hours.



## Time and cargo space

What the swap costs in time, compared with charging the same two packs on board at a quay, and the cargo slots it gives back.




Time the stop adds to the trip

**With swaps on the route: 2 packs on board**

pack pack TEU TEU TEU

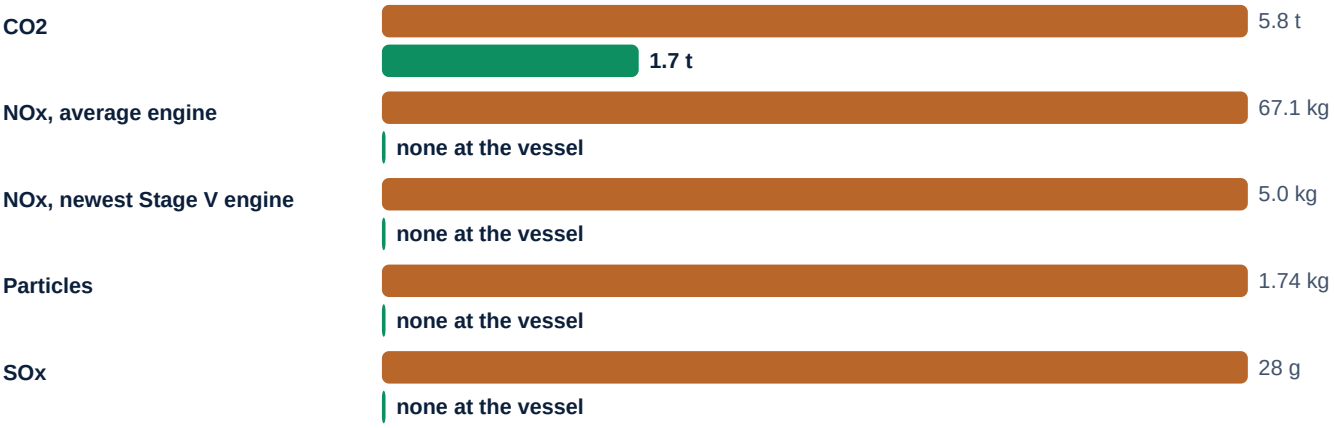
**Round trip with no swaps: 5 packs on board**

pack pack pack pack pack

Each 20 ft ZESpack takes one TEU slot. Five packs cover 176 km loaded with a 20% reserve, so carrying two gives three slots back to cargo.

## Emissions on this trip

The same trip by a diesel barge and by the e-barge charged from today's Belgian grid. On renewable supply the e-barge CO2 falls close to zero, and it emits no NOx, particles or SOx at the quay or on the canal.




CO2 for the e-barge counts the electricity at the plug, including charging and drive losses. NOx, particles and SOx for the e-barge are zero at the vessel.

Diesel barge  E-barge on swapped packs

## Three ways to swap

| Option | Time added | Equipment | Best when |
|---|---|---|---|
| **Swap at the hub** | 30 min stop | Gantry crane and docking station | When the hub is on the route and has a free slot |
| **Tender at the berth** this report | None, inside the cargo call | Battery tender with its crane | When the barge already stops for cargo |
| **Tender at the lock** | None, inside the lock wait | Battery tender with its crane | When locks are busy; even more in dry summers |

## Assumptions and sources

Loaded e-barge about 60 kWh per km; each 2.9 MWh ZESpack has 2.6 MWh usable; 20% reserve; charging and drive losses 12%. Diesel: 0.22 kg fuel per kWh and 3.17 kg CO2 per kg; NOx 8.1 g per kWh for the average engine and 0.6 g for a Stage V engine (real world measurements on the Rhine, ACP 2023); particles 0.21 g and SOx 0.37 kg per TJ (German national inventory, inland navigation). Belgian grid 182 g CO2 per kWh in 2025 (Electricity Maps). Charging two packs at a quay takes about 3 hours from a 2 MVA connection. Dry summer: group lockage when the Meuse runs low, waits of 2 to 4 hours (Wijnegem and Olen, August 2022). Real figures will replace these once the operator, terminal and grid data are in.